Module :- 6

(*) Single phase Induction motor

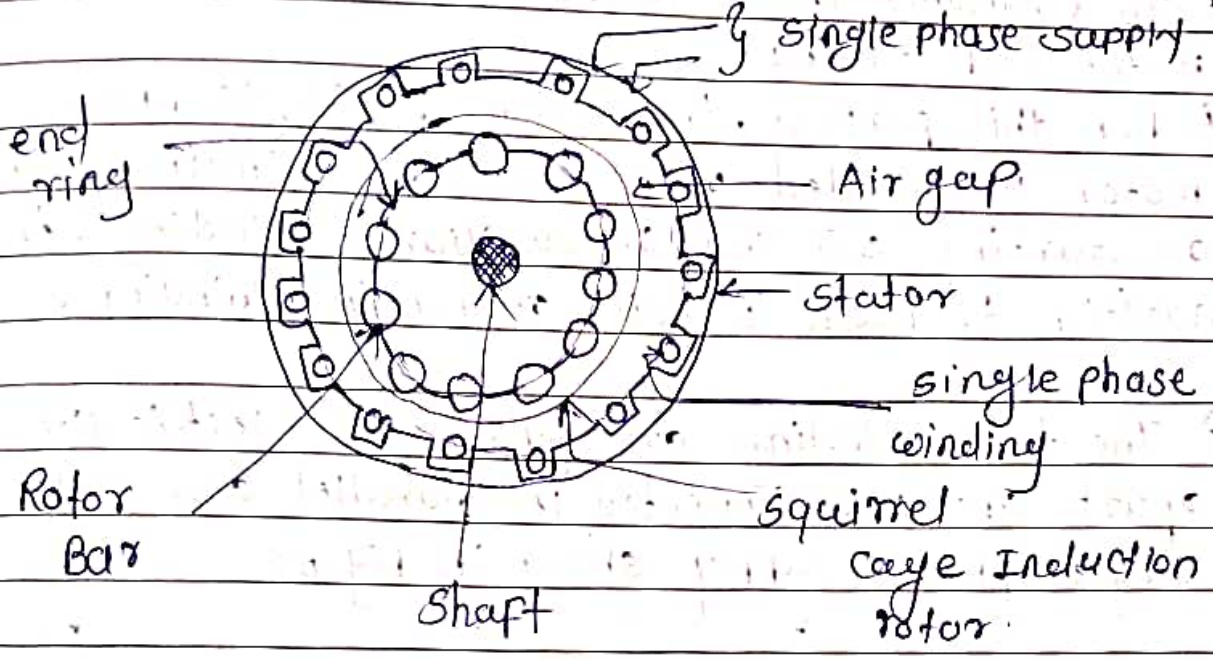


Fig a> Construction of single phase Induction motor

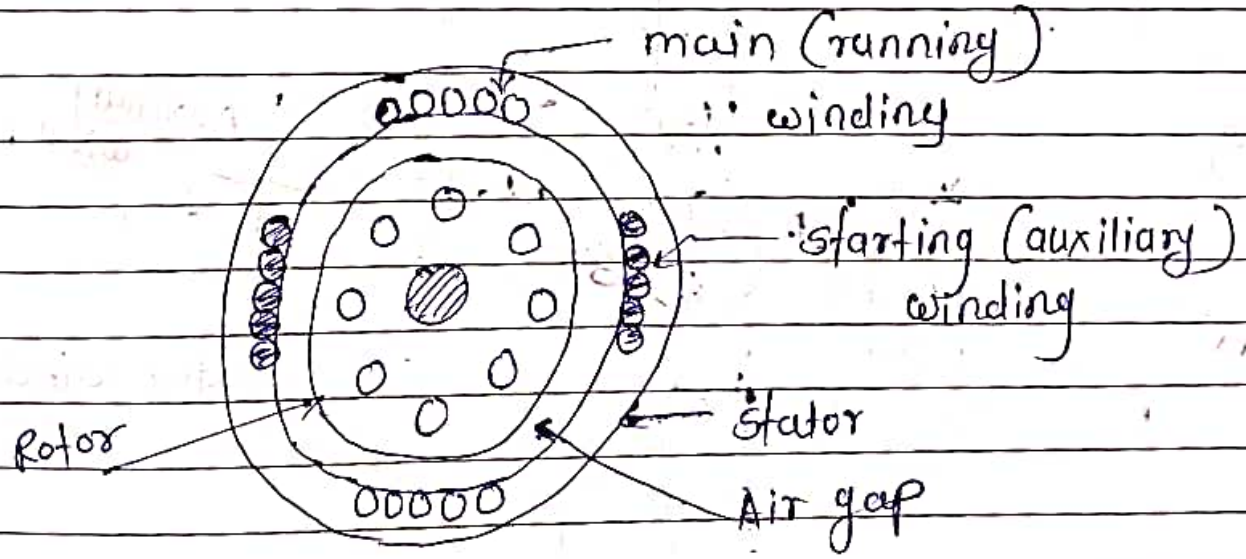


Fig b> single phase induction motor

① single phase motor has an alternating field & not a rotating field

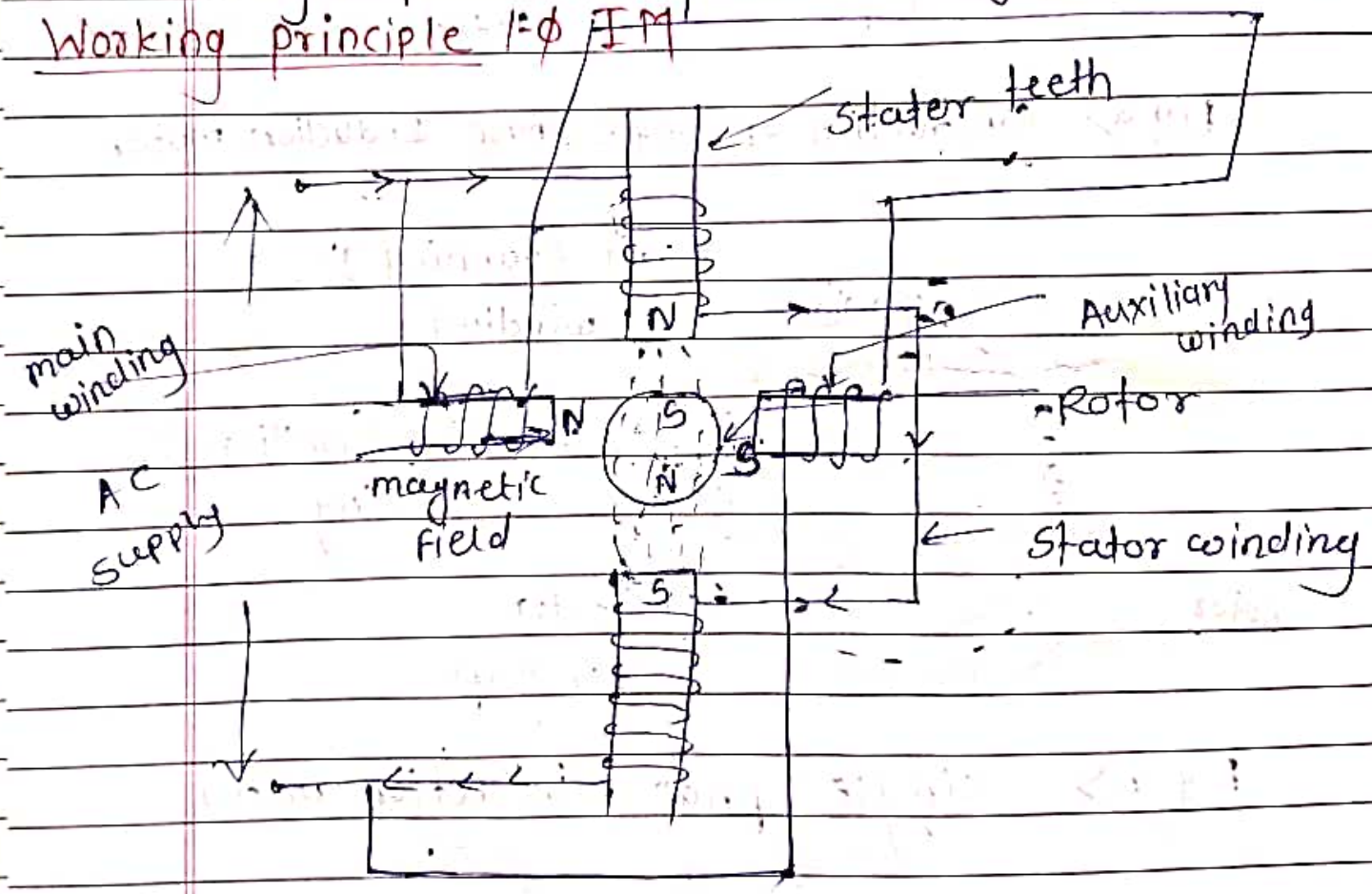
② Therefor, it is not self starting.

③ To overcome this drawback and make the motor self starting

④ for this purpose, the stator of single phase motor is provided with an extra winding, known as starting winding or auxiliary winding. in addition to main winding or running winding.

⑤ The two winding are spaced 90° electrically apart and are connected in parallel across the single phase supply shown in fig b>

Working principle 1-ϕ IM



[N] [N repuse

[N] [S attract

> The AC supply is given to stator winding, the current are start flowing through it.

> The current are flowing through the stator winding it produces the magnetic field

> The magnetic field is passing through the rotor, & The magnetic field is varying in nature & change with respect to time.

> The magnetic field is varying because the AC current is alternating & it's changes +ve half to the -ve half.

→ its a principle of Induction.

(current carrying conductor placed in magnetic field it's cut's Rotor conductor bar (magnetic field) high low

> the magnetic field is ~~increase~~ ~~decrease~~ in the rotor ~~the~~ winding & the current are generate in rotor winding

> the rotor produces it's own magnetic field

> the stator magnetic field & Rotor magnetic field is opposite to each other ~~the~~ that condition rotor is not rotating.

> that why we connect the main winding and auxiliary winding in the single phase Induction motor:

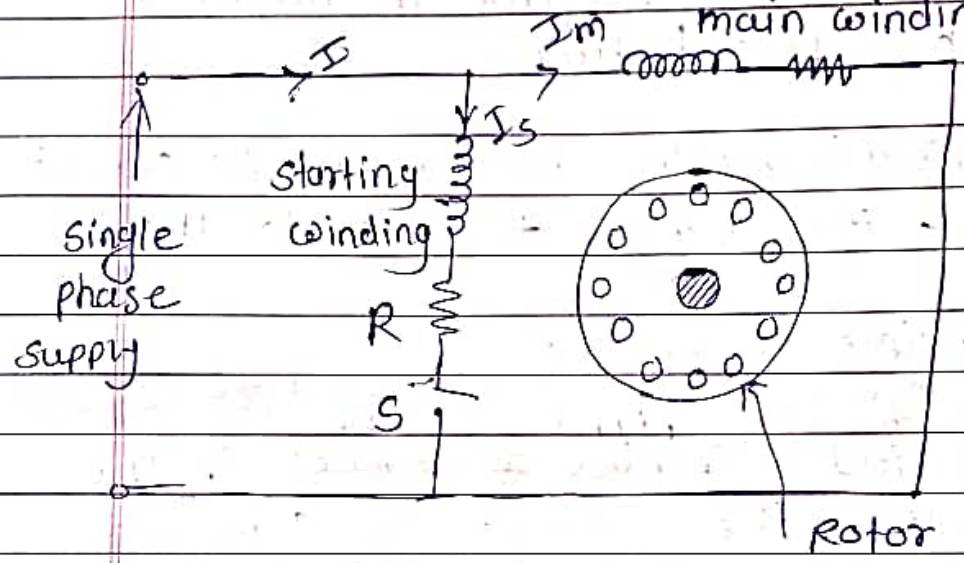
> to produces the Rotating magnetic field in Rotor & the torque are produced in to the Rotor to Rotates the Rotor.

& experiences the force to produces torque

(*) Types of single phase Induction motor

(1) ~~Slip~~ split-phase Induction motor
(2) Capacitor-start induction motor

(1) ~~Splitt~~ Split-phase Induction motor. (used to Domestic appliances)



S = Centrifugal switch

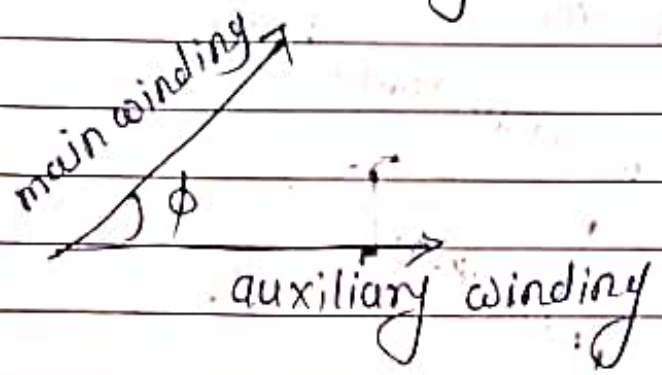
> Split phase Induction motor is the type of single phase Induction motor. that is convert electrical energy into mechanical energy

main winding $R_m \downarrow$ low & $X_m \uparrow$ high
&
starting winding $R_s \uparrow$ high & $X_s \downarrow$ low

> using the centrifugal switch, starting winding is disconnect from the ckt

> main winding & starting winding are connected in parallel to each other with the angle is 90° is space

> when we give a AC supply to the ~~to the motor~~ across the both winding the phase difference is created



> due to phase difference ~~stator~~ the rotating magnetic feild produced in the air gap between stator & rotor

> The current phase difference is less than 90° but it is enough to rotate the rotor

> rotating magnetic field cut's the Rotor bar & Ro produced magnetic field ~~in~~ across the rotor & due to interation of both magnetic field. the tangentially torque act on the rotor & rotor are rotates

> the rotor speed is ~~increases~~ about 70 to 80% percentage, the centrifugal switch are get opened to disconnect the starting winding

> & the rotor is rotates continew using main winding.

> ~~no~~ we change the terminal of main winding & ~~auxi~~ starting winding the rotor rotates in opposite direction.

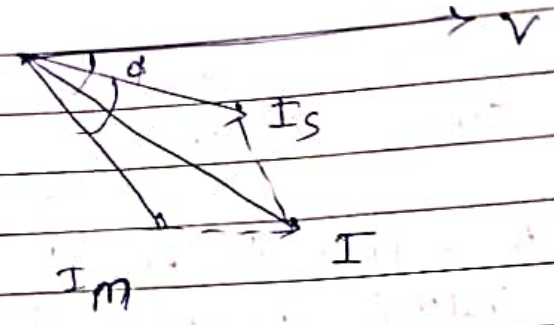


Fig b) Phasor diagram

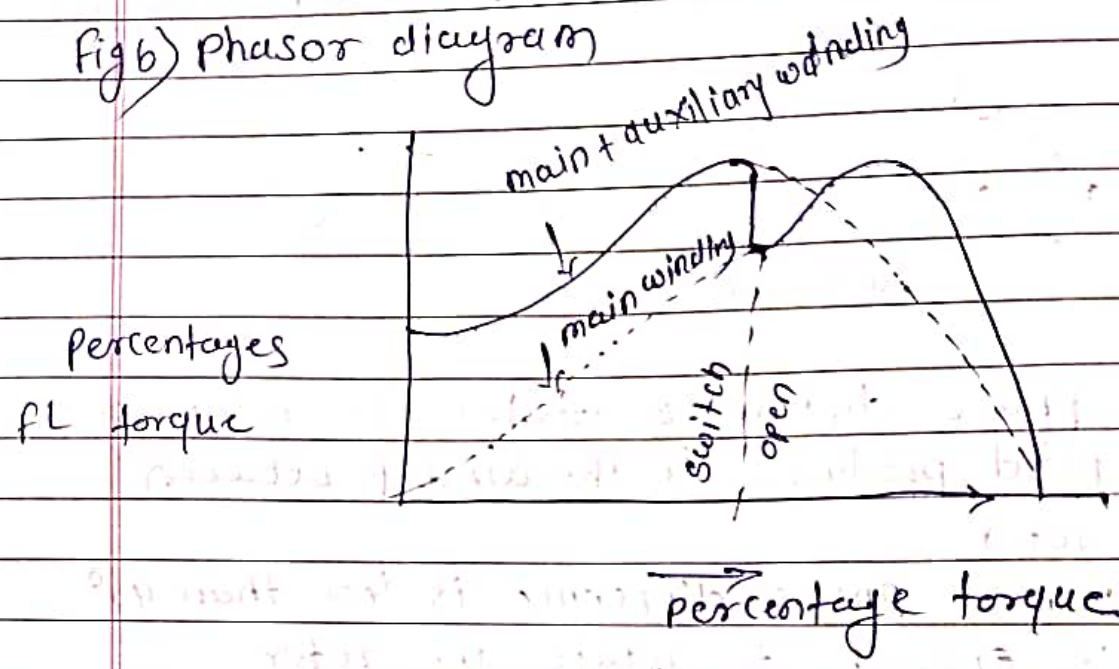
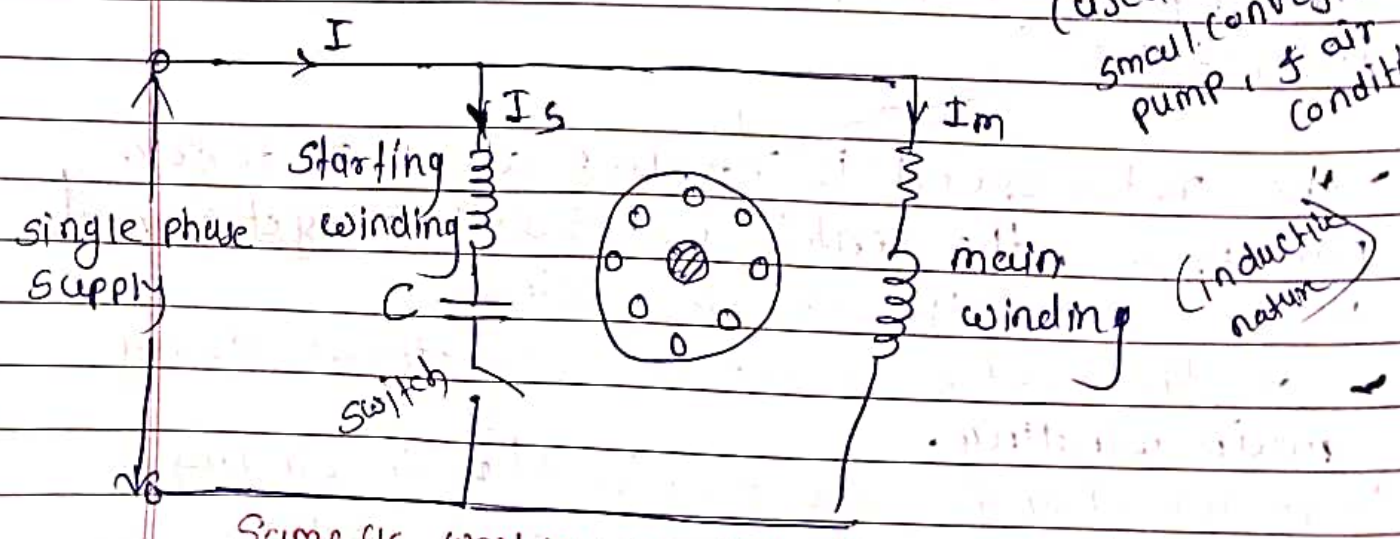


Fig c) torque-speed charactristics

② Capacitor start induction motor



Same as working for split-phase Induction motor only capacitor are connected in starting winding to start the motor & increases the speed of motor about 70 to 80%. ~~& the start~~ & this condition starting winding is disconnect from the main winding